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Exploring the potential of using an AI language model for automated essay scoring

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ABSTRACT

The widespread adoption of ChatGPT, an AI language model, has the potential to bring about significant changes to the research, teaching, and learning of foreign languages. The present study aims to leverage this technology to perform automated essay scoring (AES) and evaluate its reliability and accuracy. Specifically, we utilized the GPT-3 text-davinci-003 model to automatically score all 12,100 essays contained in the ETS Corpus of Non-Native Written English (TOEFL11) and compared these scores to benchmark levels. The study also explored the extent to which linguistic features influence AES with GPT. The results showed that AES using GPT has a certain level of accuracy and reliability and could provide valuable support for human evaluations. Furthermore, the analysis revealed that utilizing linguistic features could enhance the accuracy of the scoring. These findings suggest that AI language models, such as ChatGPT, can be effectively utilized as AES tools, potentially revolutionizing methods of writing evaluation and feedback in both research and practice. The paper concludes by discussing the practical implications of using GPT for AES and exploring prospective future considerations.

Introduction

The world was taken by surprise by the release of ChatGPT (https://chat.openai.com/) at the end of the year 2022. ChatGPT is a chatbot that answers human questions with an AI that seems to have a perfect understanding of the language. OpenAI's GPT language model had already attracted attention in the field of natural language processing (NLP), but its use was limited to users who could use computer programs such as Python. ChatGPT, however, lowers the barrier to its use by allowing anyone to ask questions on a browser and presenting answers in many languages. It is inevitable that learners and teachers will begin to use this type of tool both inside and outside of the classroom. The emergence of an AI tool capable of comprehending and producing a language like ChatGPT marks the beginning of a new era in which humans and AI coexist in L2 language learning and teaching, as well as in its research.

In this paper, we investigate the potential of using an artificial intelligence (AI) language model, specifically the GPT-3 text-davinci-003 model, for automated essay scoring (AES), particularly in terms of its accuracy and reliability. AES involves the use of computer algorithms to evaluate and provide feedback on student essays. The use of AI-based language models has gained popularity in AES due to their ability to accurately assess the quality of student writing. Our study seeks to explore the implications of using such an AI language model for AES. We also examine the potential role linguistic features could play in improving the accuracy of AES using an AI language model.

In what follows, we will provide a concise overview of AES technology, covering its historical development, advantages and disadvantages, and its practical use in high-stakes testing and as an automated written corrective feedback tool in classrooms. We

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will also review the integration of machine learning, deep learning, and transformers into AES technology. Next, we will describe the purpose and methodology of our study, and present the analysis results, including a comparison of AES scores with benchmark levels. Subsequently, we will discuss the implications of our findings, including the potential benefits and limitations of AES technology. Finally, we will conclude by exploring the possible implications of this technology for L2 researchers and practitioners.

Background

Automated Essay Scoring (AES)

Automated Essay Scoring (AES) refers to the use of technology to evaluate and score written essays. The process of AES involves using computer programs to analyze and score written work based on predefined criteria such as linguistic correctness, lexical richness, coherence, syntax, and semantic relevance. Although the use of AES has both pros and cons, it is generally recommended because it can avoid the risks associated with human scoring, such as the time-consuming and inaccurate scoring of essays. As a result, it has attracted constant attention among researchers and practitioners since its inception over 50 years ago.

The AES research can be traced back to the 1960s, with Page (1967) as the pioneering Fig. behind the Project Essay Grade (PEG). PEG utilized multiple regression analysis to predict the scores of essays based on measurable text characteristics, including the average sentence length, average essay length, number of prepositions, and number of commas, by comparing them to the scores given by human raters on similar topics. Early AES systems, such as PEG, were often criticized for solely focusing on surface structures and neglecting features pertaining to the content, making them vulnerable to cheating strategies such as adding more words and commas (Attali, 2013; Dikli, 2006).

As a result, more precise AES systems were created in the 1990s thanks to the advancement of computers. Significant progress was made in NLP, a branch of AI that uses computer programming to simulate human intellectual abilities. It allowed for more accurate scoring using not only superficial linguistic features but also deeper ones (Hussein et al., 2019). Developed by the Educational Testing Service (ETS) in 1998, e-rater is representative of such a new AES system. The e-rater system uses a new statistical and rule-based method that allows analyses of sentence structure (syntax), word structure (morphology), and meaning (semantics) (Burstein et al., 2013). As e-rater has been reported to have high reliability and validity (Attali & Burstein, 2004), it has been officially used in combination with human raters in high-stakes tests such as the Graduate Record Examination (GRE) and Test of English as a Foreign Language (TOEFL).

At the same time, because the AES systems come equipped with two engines, one is for calculating scores and the other for providing automatic feedback on errors, there has been much research, under the umbrella term—Automated Writing Evaluation (AWE), on using the feedback function as automated written corrective feedback for student writing in the classroom (e.g., Cotos, 2014; Koltovskaia, 2020; Ranalli, 2013). For example, e-rater has a web-based AWE tool called Criterion, and IntelliMetric, another AES system, has an AWE tool, MY Access!, and the effectiveness of those AWE tools has been researched in the classroom settings (e.g., Chen & Cheng, 2008; Dikli & Bleyle, 2014; Z. Li et al., 2014). Some researchers are skeptical about using those AES-based AWE tools because the writing constructs those tools measure are different from what is valued in the classroom (Condon, 2013). However, others argue that AES-based AWE tools such as Criterion help learners increase revisions and accuracy and, as a result, have a positive impact on the quality of texts (J. Li et al., 2015; Stevenson & Phakiti, 2014).

Machine learning and deep learning

The field of AES research has vastly benefitted from the development of the machine learning method, a sub-discipline of AI, with which it is possible to construct different automated scoring models and cross-validate them. Machine learning in an AES system works by training a model on a large corpus of essays scored by human expert raters. The model then uses the patterns it learned from this data to automatically predict the score of new, unseen essays. The ultimate goal is for the model to make accurate predictions that align with the scores given by human raters. Typically, the model is trained on various features of the text, and as such, selecting critical features is the pivotal step in machine learning (Mizumoto, 2023) because it will yield better results and reduce the risk of using irrelevant or noisy features, which can negatively impact the performance of the model. As can be expected, machine learning is known to work better than other methods in AES (Shermis & Burstein, 2003).

Machine learning techniques have been introduced to the L2 field in the previous decade (Crossley, 2013) along with freely accessible NLP tools that enable users to automatically extract a variety of linguistic features such as text cohesion (Crossley et al., 2016), syntactic complexity (Kyle & Crossley, 2018; Lu, 2010), syntactic sophistication (Kyle & Crossley, 2017), lexical sophistication (Kyle & Crossley, 2014), lexical complexity (Lu, 2012; Spring & Johnson, 2022), and lexical diversity (Kyle et al., 2021). As these linguistic features have been reported to show a positive correlation with human-rated scores, by applying machine learning algorithms (e.g., support vector machine, random forest, and neural networks) and selecting key features, a stronger predictive model can be built, with which it is possible to better understand writing quality and the development of L2 learners (Crossley, 2020).

In recent years, state-of-the-art in AES has evolved by incorporating deep learning. Deep learning refers to a subfield of machine learning that utilizes very huge, multiple hidden layers of neural networks that model and solve complex problems, such as image and speech recognition, NLP, and decision making by using massive amounts of structured and unstructured data. Although deep learning is categorized as part of machine learning, the difference between deep learning and other machine learning techniques is that deep learning uses multiple layers (like a sequence of steps in solving a puzzle) to learn and make decisions, whereas other machine learning techniques typically use a single layer. Thus, deep learning performs tasks better with high accuracy and generalization ability than other machine learning algorithms do, and it is what is behind the great improvements in the accuracy of neural machine translation (NMT) tools such as DeepL and Google Translate (Rivera-Trigueros, 2022). While deep learning extracts features directly from raw data by itself, features are often handcrafted by domain expert humans (e.g., researchers) in other machine learning techniques.

With these important technological developments, modern AES makes use of deep learning approaches so that the model can take the content of the essay, inducing syntactic and semantic features, into account, in addition to the surface features (e.g., Dong et al., 2017; Hussein et al., 2019; Taghipour & Ng, 2016). Shin and Gierl (2021) compared two algorithms: (a) support vector machines (SVMs) in conjunction with Coh-Metrix features as a traditional AES model and (b) convolutional neural networks (CNNs) approach as a more contemporary deep-neural model. They reported that the CNNs model performed better, with the results more comparable to the human raters than the traditional model. As witnessed in this example, it has been widely recognized that applying deep learning (i.e., neural network) approaches to AES yields better results than earlier approaches. In fact, in a systematic literature review, Ramesh and Sanampudi (2022) highlighted that most AES systems developed recently utilize the concept of neural networks.

Transformers

Among the deep learning approaches, the one that has had the greatest impact and will undoubtedly be key in the future of NLP is called “transformers.” Transformers are a type of deep learning neural network architecture designed to learn context and meaning from sequential data (Vaswani et al., 2017). Transformers are an evolution or extension of previous neural network architectures: convolutional neural networks (CNNs) and recurrent neural networks (RNNs) by combining the benefits of them (Giacaglia, 2019). Transformers can be trained faster and more efficiently and still achieve better results than other models; therefore, they have been instrumental in many recent breakthroughs in NLP tasks that turn an input sequence into an output sequence.

BERT (Bidirectional Encoder Representations from Transformers) is one of the most well-known transformer-based large language models. Pre-trained on a large amount of text data, BERT was developed by researchers at Google AI Language (Devlin et al., 2018). The advantage of BERT is its ability to understand the context of a word based on the surrounding words in a sentence. It has been used in a wide range of NLP tasks such as sentiment analysis, named entity (e.g., people, organizations, and locations) recognition, auto summarization, and question answering. BERT has been utilized in the field of applied linguistics as well. For example, by using BERT, Lu and Hu (2022) showed that two of the context-aware lexical sophistication measures created with contextualized word embeddings using the BERT model correlated more strongly with L2 English writing quality than traditional lexical sophistication measures obtained with TAALES (Crossley et al., 2018). Lu and Hu also showed that incorporating such context-aware lexical sophistication measures into the regression model, in addition to the traditional ones, improved the model’s predictive power in accounting for L2 English writing quality. Similar results have been reported in the AES research. While many studies suggest that BERT outperforms other models in AES performance, combining BERT with handcrafted features achieves the current state-of-the-art AES performance (Lagakis & Demetriadis, 2021).

GPT (Generative Pre-trained Transformer) is another groundbreaking transformer-based language model (Radford et al., 2018) developed by OpenAI (https://openai.com/). GPT has been fine-tuned by using a massive corpus of text data for many NLP tasks, making it capable of generating coherent and fluent text outputs such as text generation, language translation, and question-answering in a human-like manner. BERT and GPT differ in that BERT is a bidirectional transformer-based architecture, while GPT is a unidirectional one. That is, BERT looks at a text from both the beginning and the end, whereas GPT only looks at the text from start to end. BERT was trained using a masked language-modeling objective (i.e., fill in missing words in sentences), while GPT was trained on a much larger corpus than the one for BERT to be able to generate answers to questions. OpenAI has released a series of GPT models: GPT-1 (in 2018), GPT-2 (in 2019), and GPT-3 (in 2020) with different sizes and capabilities, and GPT-3 is much larger than BERT, as it has 175 billion parameters (numbers used by the program to help it learn and understand language) compared to BERT’s 340 million parameters. With a different architecture and training approach (bidirectional and masked language-modeling objective for BERT, versus unidirectional and large corpus of text for GPT), BERT is typically used for natural language understanding tasks, whereas GPT is for natural language generation tasks (e.g., C. Li & Xing, 2021).

Although GPT has shown exceptional results in various language generation tasks, it has yet to be utilized in AES. This may be due to the fact that GPT primarily functions as a language generation model and is therefore deemed inappropriate for AES tasks. Conversely, BERT has been applied to AES, as mentioned above, because it has proven to be efficient in tasks such as sentiment analysis and text classification, which are similar to AES. Additionally, while GPT is capable of performing a diverse range of tasks effectively when given suitable prompts through its prompt-based learning approach, it cannot normally undergo task-specific fine-tuning. Task-specific fine-tuning is often crucial for enhancing performance in complex tasks such as AES as they necessitate a deeper understanding and representation of the language.

Despite these limitations, the possibility of using GPT for AES should not be disregarded completely. Mayer et al. (2023) conducted a study on the application of GPT in the classification of business e-mail replies as polite or impolite, which is comparable to AES. The results of the study indicate that GPT, utilizing a prompt-based approach that eliminates the need for programming, can attain accuracy levels similar to human ratings. Mayer et al. posit that GPT-based machine ratings can serve as a supplementary tool rather than a substitute for human ratings. The prompt-based language approach of GPT provides a user-friendly interface, facilitating the engagement of non-experts in AES. For this reason, Mayer et al. suggest that “this new, more intuitive, prompt-based learning approach will enable more people to use artificial intelligence” (Mayer et al., 2023, p. 125). Furthermore, since the release of ChatGPT in late November 2022, the accessibility of GPT technology has significantly ehnahnced. ChatGPT is a web-based platform that is available to virtually everyone, except for some countries, and as of February 2023, it is free to use. This development has led to suggestions that ChatGPT could be utilized for AES applications (e.g., Essel, 2023).

The present study

If GPT can be utilized for AES, it has the potential to increase the accessibility of the AES technology and make it more widely used in L2 research and practice. However, there has not been any research that applies GPT to AES to date. Scoring consistency is an important aspect of evaluating the AES system, and this refers to whether and to what extent AES by GPT can reproduce scores by human raters (Enright & Quinlan, 2010). Relatedly, given the accumulated research on linguistic features in L2 writing research (Crossley, 2013), an important question to consider is whether AES by GPT outperforms research-based linguistic features in predicting the scores given by human raters. Given these backgrounds, this study aimed to explore the potential use of prompt-based GPT, similar to ChatGPT, for AES. Following are the two research questions that were posed for this study.

RQ1: To what extent is AES by GPT reliable?
RQ2: To what extent does AES by GPT predict benchmark essay scores relative to a set of research-based linguistic features?

Methods

Dataset

In this study, we used ETS Corpus of Non-Native Written English, also known as TOEFL11 (Blanchard et al., 2013) as a source of essays written by English learners. TOEFL11 was developed by Educational Testing Service (ETS) and distributed via the Linguistic Data Consortium (https://catalog.ldc.upenn.edu/LDC2014T06). TOEFL11 is a collection of 12,100 English essays written by individuals who took the TOEFL test in 2006 and 2007 and who had 11 distinct native languages (as reflected in the name, TOEFL11): Arabic, Chinese, French, German, Hindi, Italian, Japanese, Korean, Spanish, Telugu, and Turkish. The corpus is composed of data obtained through random selection, and as a result, it encompasses 1,100 essays per language for a total of 12,100 (i.e., 1,100 × 11), which are evenly selected from eight given prompts. Each essay has been assigned one of the three score levels: Low, Medium, or High. Given its meticulous design and the availability of benchmark scores, TOEFL11 was deemed the most appropriate source for AES by GPT in this study.

Type of GPT

We utilized, in this study, the OpenAI's text-davinci-003 model, which was released on November 30th, 2022 as part of the GPT-3.5 series. It is noteworthy that the text-davinci-003 model and ChatGPT, which was also released on the same day, are both members of the GPT-3.5 series. However, ChatGPT has undergone additional training utilizing the Reinforcement Learning from Human Feedback (RLHF) technique to address misalignment issues and improve its functionality as a chatbot. Given that the underlying model is the same, the performance of both models can be considered comparable, allowing us to extend the conclusions of this study to the use of ChatGPT. This also is the reason why we did not engage in further fine-tuning in this study. In other words, our findings should be applicable to future studies using ChatGPT or similar models that do not permit users to fine-tune the GPT language model on a specific task.

Prompt

In addition to the prompt which orders the text-davinci-003 model to score a given essay, the IELTS TASK 2 Writing band descriptors (public version) were selected as the rubric included in the prompt for this study, as they provide a detailed evaluation of (a) task response, (b) coherence and cohesion, (c) lexical resource, and (d) grammatical range and accuracy through a 10-point score (band) scale from 0 to 9. This scoring rubric was preferred over the 5-point scale used in the TOEFL independent writing scoring rubric, as the 10-point scale would offer a more nuanced evaluation and enables greater differentiation among scores. We did not include sample essays with specific scores in the prompt. The example of the actual prompt used in the study is depicted in Fig. 1.

We automatically scored each essay with the text-davinci-003 text completion API from OpenAI. API requests were sent with parameters and the prompt design using Python 3.8.5. The cost for using text-davinci-003 through OpenAI's API was $0.02 per 1,000 tokens.

Stratified sampling

After scoring all 12,100 essays using the procedure described above, we randomly selected a stratified sample of 1,210 essays for re-scoring to assess the consistency of the initial scoring. The sample was designed to represent the distribution of three score levels: Low, Medium, and High, with 110 essays selected randomly from each of the 11 first-language test-takers. We re-scored the 1,210 sampled essays using the same prompt, and it was found that the mean scores of the three levels in the randomly sampled essays had no statistically significant differences compared to those of the three levels in the original 12,100 essays. All confidence intervals (CIs) of effect sizes included zero (see the online supplementary material for details).

```
I would like you to mark an essay written by English
as a foreign language (EFL) learners. Each essay is
assigned a rating of 0 to 9, with 9 being the
highest and 0 the lowest. You don't have to explain
why you assign that specific score. Just report a
score only. The essay is scored based on the
following rubric.
```

[*IELTS rubric in a plain text format.*]

```
ESSAY:
```

[*Inserting each of the 12,100 essays using a for loop in Python code.*]

Fig. 1. Example of the Prompt Used in the Study
Note. See the online supplementary material for the actual code and rubric.

Linguistic Features

Following previous research on linguistic correlates of human rating scores, we considered a range of linguistic features at the levels of lexis, phraseology, syntax, and cohesion. Table 1 lists a total of 45 research-based linguistic measures considered in the current study. We selected these measures based on previous studies in each domain, which demonstrated utilities of the features in each of the following domains: lexical diversity (Kyle et al., 2021; McCarthy & Jarvis, 2010; Zenker & Kyle, 2021), lexical sophistication (Crossley et al., 2018), syntactic complexity (e.g., Lu, 2010; Wolfe-Quintero et al., 1998), fine-grained syntactic features (Kyle & Crossley, 2018), verb-argument construction measures (Kyle & Crossley, 2017), and textual cohesion measures (e.g., Crossley et al., 2019). The measures were automatically computed using the tools indicated in the rightmost column in Table 1.

Statistical analysis

To address Research Question 1, we analyzed the scores of all 12,100 essays based on three levels: Low, Medium, and High, and calculated descriptive statistics for each level. Then, we used inferential statistics and effect sizes to test for mean differences among the levels. To assess intra-rater reliability, we utilized the Quadratic Weighted Kappa, which is a variant of Cohen's Kappa, on a stratified random sample of 1,210 essays that were re-scored by the GPT model. This allowed us to determine the level of agreement between the first and second scores of these essays, while accounting for the possibility of agreement occurring by chance.

To answer Research Question 2, we used a series of single-level ordinal regressions to assess the ability of AES by GPT to reproduce the benchmark levels of the stratified random sample comprising 1,210 essays. Our key focus was to determine whether the integration of linguistic measures could enhance the precision of AES. To do this, we compared the performance of multiple regression models using a model comparison approach. These models included:

Model 1: GPT scores only
Model 2: GPT scores + Lexical measures
Model 3: GPT scores + Syntactic complexity measures
Model 4: GPT scores + Fine-grained syntactic dependency + Verb argument construction
Model 5: GPT scores + Cohesion measures
Model 6: GPT scores + All linguistic measures above
Model 7: Linguistic measures only

If AES by GPT is comparable to the linguistic feature model, we would expect Models 1 and 7 to perform similarly. On the other hand, if the inclusion of linguistic features does improve prediction accuracy, then Models 2-6 should outperform Model 1.

We constructed each regression model through the Bayesian approach and compared their model fit through information criteria, specifically the Leave-One-Out cross-validation (LOO) information criterion (IC) (Vehtari et al., 2017). LOOIC considers both the model fit to the data and the complexity of the model, avoiding overfit. This model comparison approach is preferred to the variable selection approach using step-wise regression because purely statistical variable selection ignores the theoretical implication of the model (McElreath, 2020). One advantage of Bayesian estimation in this study is that the uncertainty of Information Criteria can be computed, enabling us to more objectively evaluate the relative improvement of the model fit. We fitted each regression model using the *brms* package (Bürkner, 2017), and the LOOIC was computed using the *loo* package (Vehtari et al., 2022). For each model, a weakly informative prior (with student-t distribution with mean = 0, SD = 1, and degrees of freedom = 3) was employed to estimate the standardized regression coefficient (by the z-score transformation of each predictor variable). The robustness of the models against the prior selection was checked by refitting the same models with varying SDs (i.e., 2.5 and 5, respectively). The convergence of the

Table 1
Selected Linguistic Features Used in the Study

Areas	Linguistic Features	Tools
Lexical Diversity		
	Moving Average TTR (50 words) All words	TAALED
	Measure of Textual Lexical Diversity (original) All words	TAALED
Lexical Sophistication		
	Kuperman Age of Acquisition Content Words	TAALES
	Brysbaert Concreteness Content Words	TAALES
	COCA Academic Frequency Log Content Words	TAALES
	COCA Academic Frequency Log Function Words	TAALES
	COCA Academic Bigrams Mutual Information	TAALES
	COCA Academic Trigrams Mutual Information	TAALES
	COCA Academic Trigrams Proportion (30k)	TAALES
	McDonald Contextual Diversity Content Words	TAALES
	ALL Academic Word List	TAALES
	ALL Academic Formulas List	TAALES
Syntactic Complexity Measures		
	Mean Lengths of Clause	SCA (using TAASSC)
	Verb Phrases per T-unit	SCA (using TAASSC)
	Clauses per T-unit	SCA (using TAASSC)
	Dependent Clauses per T-unit	SCA (using TAASSC)
	Complex Nominals per Clause	SCA (using TAASSC)
Fine-grained Syntactic Dependency		
	Dependents per Nominal	TAASSC
	Dependents per Nominal (Standard Deviation)	TAASSC
	Adjective Complements per Clause	TAASSC
	Adverbial Clauses per Clause	TAASSC
	Clausal Complements per Clause	TAASSC
	Clausal Subjects per Clause	TAASSC
	Nominal Subjects per Clause	TAASSC
	Passive Nominal Subjects per Clause	TAASSC
	Prepositions per Clause	TAASSC
	Open Clausal Complements per Clause	TAASSC
	Adverbial Modifiers per Clause	TAASSC
	Modal Auxiliaries per Clause	TAASSC
Verb-argument Construction Measures		
	Average Lemma Construction Combination Frequency, Log Transformed - All	TAASSC
	Average Delta *p* Score Verb (Cue) - Construction (Outcome) (Types Only) - All	TAASSC
	Average Delta p Score Construction (Cue) - Verb (Outcome) (Types Only) - All	TAASSC
Cohesion Measures		
	Synonym Overlap Paragraph (Noun)	TAACO
	Synonym Overlap Paragraph (Verb)	TAACO
	Word2vec Similarity All (Paragraph) 1	TAACO
	Word2vec Similarity All (Paragraph) 2	TAACO
	Basic Connectives	TAACO
	Conjunctions	TAACO
	Addition	TAACO
	Sentence Linking	TAACO
	Order	TAACO
	Reason and Purpose	TAACO
	All Causal	TAACO
	Positive Causal	TAACO
	Opposition	TAACO

Note. TAALED: Tool for the Automatic Analysis of Lexical Diversity, TAALES: Tool for the Automatic Analysis of Lexical Sophistication, TAASSC: Tool for the Automatic Analysis of Syntactic Sophistication and Complexity, TAACO: Tool for the Automatic Analysis of Cohesion. All the tools are freely available on the website “NLP for the Social Sciences” (https://www.linguisticanalysistools.org/).

model was assessed by inspecting the *R*-hat values (≤ 1.01) as well as visual inspections of the trace plot (Vehtari et al., 2021). As measures of effect size, we computed the Quadratic Weighted Kappa and their frequentist confidence intervals between the original TOEFL11 levels and the predicted levels based on each regression model. We also computed McKelvey and Zavoina’s (1975) pseudo R^2 for generalized linear models using the following formula:

$$Pseudo - \mathrm{R}^2 = \frac{Explained\ variance}{Explained\ variance\ +\ Residual\ variance}$$

where the residual variance is assumed to follow $\pi^2/3$ in logistic models (McKelvey & Zavoina, 1975). Following Gelman et al. (2019), we computed the posterior distributions of pseudo R^2 for each of the seven models.

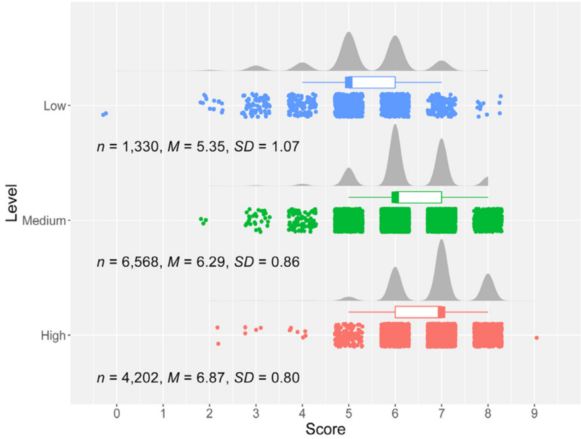


Fig. 2. Distribution of Scores Given by GPT to Each Level of TOEFL11

Table 2
Confusion Matrix of the First and Second AES by GPT

Score		The score obtained in the second AES using GPT								
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
The score obtained in the first AES using GPT	**1**	0	0	0	0	0	0	0	0	0
	2	0	1	0	0	0	0	0	0	0
	3	0	1	3	4	2	0	0	0	0
	4	0	0	3	8	10	6	0	0	0
	5	0	0	3	10	73	63	13	0	0
	6	0	0	0	3	60	246	123	13	0
	7	0	0	0	0	7	141	235	57	0
	8	0	0	0	0	0	5	55	64	1
	9	0	0	0	0	0	0	0	0	0

Note. The numbers surrounded by squares on the diagonal line indicate the exact agreement between the first and second AES by GPT.

All statistical analysis was executed using R version 4.1.2 (R Core Team, 2021). To ensure the reproducibility and transparency of the data analysis process, the data (excluding the raw data from TOEFL11), the prompt, and the Python and R code used in the study, have been made accessible on IRIS (https://www.iris-database.org/details/5FEK8-eJ6Wq) and OSF (https://osf.io/pf564/).

Results

The results of the automated scoring of essays in TOEFL11 using the text-davinci-003 GPT model are presented in Fig. 2. A clear trend was observed in the score distributions across the three levels (Low, Medium, and High), with the most frequent score being 5 for the Low level, 6 for the Medium level, and 7 for the High level. This was further supported by the median scores shown in the box plots, which are in line with these values. The mean scores of the three groups were found to be statistically different, with a medium to large effect size according to the benchmarks of small (d = .40), medium (d = .70), and large (d = 1.00) effects in L2 studies (Plonsky & Oswald, 2014). The effect size (d) for the comparison between the low and medium groups was 1.06, with a 95% CI [1.00, 1.12]. The effect size for the comparison between the low and high groups was 1.74, with a 95% CI [1.67, 1.81]. The effect size for the comparison between the medium and high groups was 0.68, with a 95% CI [0.64, 0.72]. Due to the non-normal distribution of scores in each group, non-parametric tests were also conducted, yielding consistent results.

A notable difference in means and rankings among the three levels was observed. However, differentiating the groups based solely on the scores generated by the text-davinci-003 GPT model proved difficult due to the extensive overlap of overlapping scores within a range of 4 to 8 points across the three levels. Indeed, when the groups were categorized as Low (≤ score 5), Medium (= score 6), and High (≥ score 7), following the most frequent distribution in each group, the match (i.e., exact agreement) with the original TOEFL11 level was 54.33%, which was rather low. However, taking into account the potential 1-to-2-point variation (which will be discussed later) in the scoring by the text-davinci-003 GPT model, when the groups were categorized as Low (≤ score 5), Medium (= score 5-7), and High (≥ score 6), allowing for overlapping scores and compared with the original level, the match (i.e., adjacent agreement) was 89.15%. This result suggests that the GPT scoring follows the same general pattern as the benchmark TOEFL11 level, which implies the feasibility of using GPT for AES.

The confusion matrix of the first and second AES by the text-davinci-003 GPT model, as shown in Table 2, was calculated using a stratified random sample of 1,210 essays re-scored by the GPT model. Of the 1,210 essays, there were 630 cases (52.07%) of exact

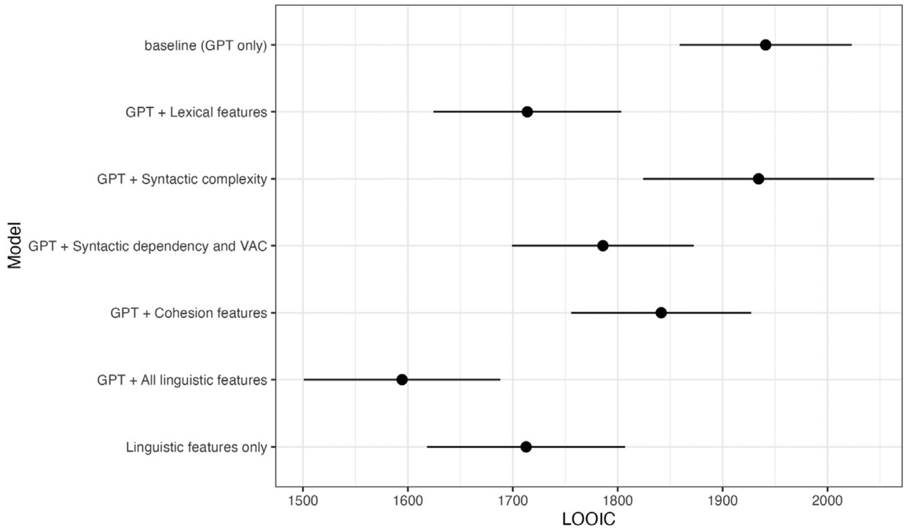


Fig. 3. LOOIC and Associated Uncertainties for Model Comparison
Note. LOOIC stands for Leave-One-Out Information Criterion. A lower LOOIC indicates a better fit and a higher predictive accuracy.

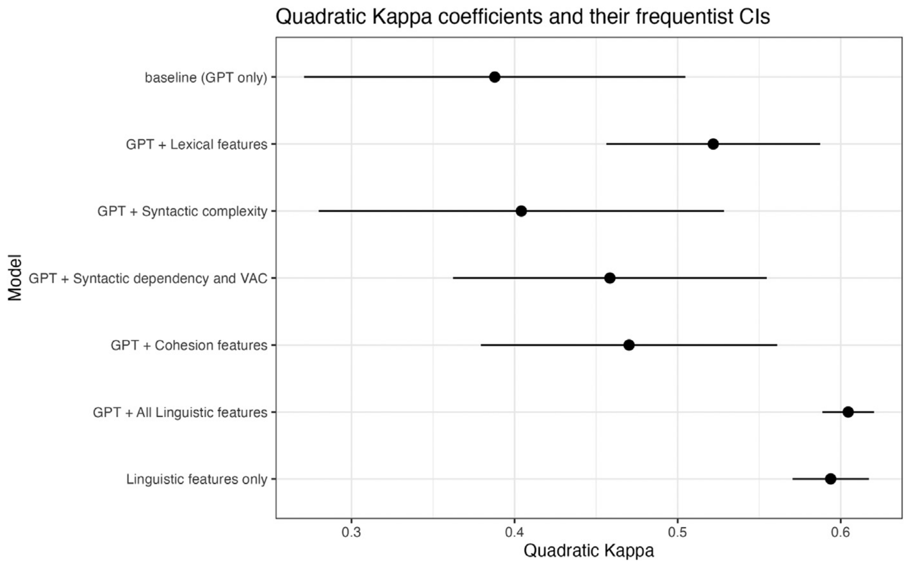


Fig. 4. The Quadratic Weighted Kappa Coefficients and Their Frequentist CIs

agreement between the first and second scores, 528 cases (44%) of a difference of one point, and 52 cases (4%) of a difference of two points. There were no cases with a difference of three points. The Quadratic Weighted Kappa was .682, 95% CI [.626, .738], indicating "substantial" agreement according to the Landis and Koch (1977) guidelines for interpreting Kappa values.

To examine the extent to which AES by GPT is able to produce essay scores that are consistent with those of the benchmark levels of TOEFL11 (i.e., professional human ratings) relative to linguistic features, seven competing regression models were constructed. Fig. 3 shows the relative model fits, with lower LOOIC values indicating better model fits.

Our results indicate that three regression models outperformed the baseline GPT regression model: (1) GPT + All the linguistic features, (2) Linguistic Features only model, and (3) GPT + Lexical feature model. These findings suggest that the benchmark levels of TOEFL11 were best predicted when GPT scores were combined with a range of linguistic features, including lexical, syntactic, and cohesion features. It is noteworthy that the linguistic features only model performed equally well as the GPT + Lexical feature model. Additionally, the results indicated that lexical features had the most impact on supplementing the AES by GPT, followed by fine-grained syntactic features and cohesion features. A similar pattern emerged when the Quadratic Weighted Kappa was computed between the benchmark levels of TOEFL11 (i.e., High, Medium, and Low) and the predicted rating based on the ordinal regression model (Fig. 4). In particular, the Quadratic Kappa Coefficient for the baseline GPT-only model was only fair to moderate (.388, 95%CI

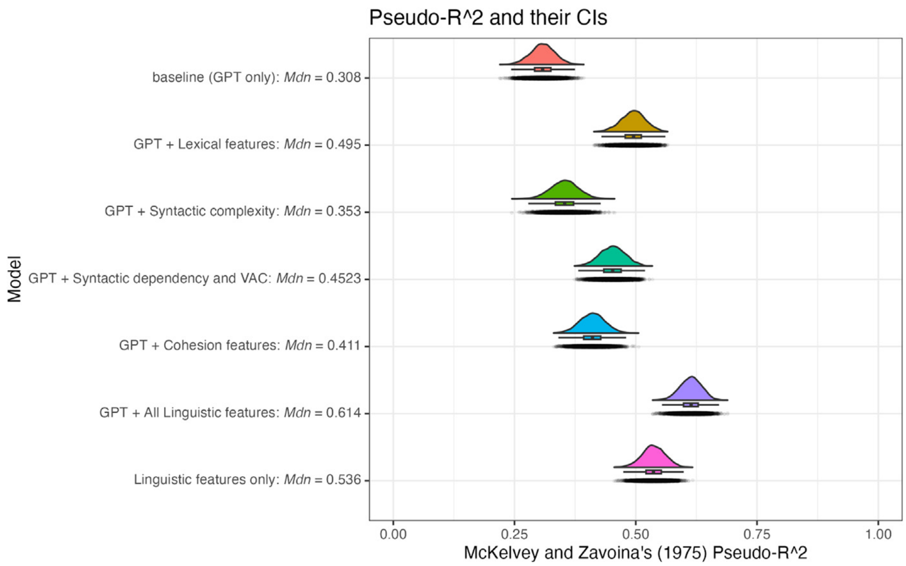


Fig. 5. The Posterior Distributions of McKelvey and Zavoina's (1975) Pseudo R^2

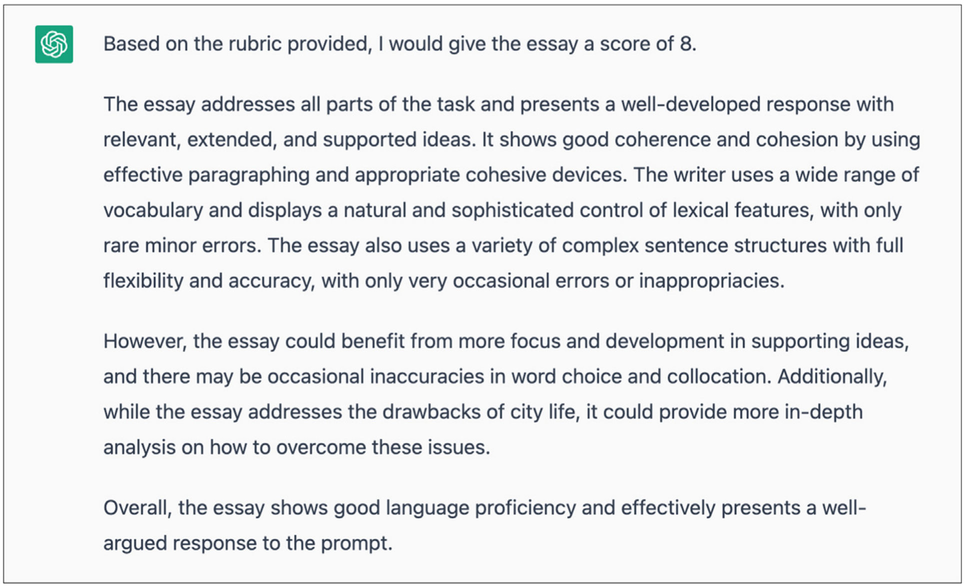
[.271, .505]), according to the Landis and Koch (1977) guidelines. Additions of linguistic measures significantly improved the kappa coefficient (with no overlaps between the two sets of CIs), reaching .605 (95% CI [.589, .620]), or "substantial" agreement. The finding was also corroborated by the analysis based on McKelvey and Zavoina's (1975) Pseudo R^2 (Fig. 5).

Discussion

The purpose of this study was to explore the potential use of GPT in automated essay scoring (AES). We also investigated the extent to which AES by GPT predicts the benchmark levels of TOEFL11 with and without the help of research-based linguistic features at the levels of lexis, syntax, and cohesion. The first research question aimed to determine the level of reliability of AES using GPT. The analysis results showed that although there was some variation in the scoring of 1-2 points (mostly 1 point), it was found that it reflects the three writing levels of TOEFL11, which was the gold standard in this study. This result can be considered to support the idea that AES using GPT can be carried out with some degree of accuracy. This suggests that AES using GPT language models such as ChatGPT could be utilized in research and practice. The second research question examined whether and how AES by GPT predicts the benchmark levels of TOEFL11 with and without other linguistic features. It has been found that additions of linguistic features significantly improved the prediction of benchmark levels, indicating that the combination of GPT + research-based linguistic features may produce the best result in predicting professional ratings.

As has been noted in the literature, the process of human rating of essays can be fraught with issues that can lead to unreliability. Factors such as fatigue, subjectivity, and inconsistency can all play a role in producing unreliable results (Hussein et al., 2019). Furthermore, the process of human rating is typically time-consuming and labor-intensive. Given these challenges, the utilization of AES tools leveraging GPT can offer valuable support in a range of settings. Employing AES with GPT has several benefits, including shorter rating times and increased consistency in scoring. As such, this approach is an attractive alternative for teachers and researchers. The use of AES with GPT has the potential to revolutionize writing instruction and assessment, providing immediate scoring and augmented feedback on writing quality that aligns with specific criteria. For example, Fig. 6 provides an illustration of the type of feedback that ChatGPT can provide when scoring an essay based on the rubric utilized in this study. In this example (Fig. 6), the essay received a score of 8. If a request is made to ChatGPT to revise the essay to achieve a score of 9, the system will provide a modified version that maintains the original content while improving its quality. The revised essay can serve as a model essay, which has been shown to enhance learner noticing (Nguyen & Le, 2022). (However, it is important to note that unethical conduct, such as presenting the revised essay as one's own work, is of course not acceptable.) This feedback method can effectively reduce the workload of teachers in correcting student writing, mitigate evaluation bias, and allow teachers to concentrate on other crucial aspects of writing instruction such as overall structure, coherence, content, and writing strategies.

AES with GPT can also be utilized as an Automated Writing Evaluation (AWE). This means that it can provide detailed feedback to both learners and teachers, in addition to giving scores. For example, if the prompt given to GPT is "*Please correct errors in this essay and provide the original sentence and the revised sentence in two columns*," it can display the original and corrected sentences in a two-column table format for detailed feedback. Additionally, by using a prompt such as "*Create a table that presents the original sentences, corrections, and specific reasons in an easy-to-understand format*," it is possible to include the reasons for the errors in a third column.



Based on the rubric provided, I would give the essay a score of 8.

The essay addresses all parts of the task and presents a well-developed response with relevant, extended, and supported ideas. It shows good coherence and cohesion by using effective paragraphing and appropriate cohesive devices. The writer uses a wide range of vocabulary and displays a natural and sophisticated control of lexical features, with only rare minor errors. The essay also uses a variety of complex sentence structures with full flexibility and accuracy, with only very occasional errors or inappropriacies.

However, the essay could benefit from more focus and development in supporting ideas, and there may be occasional inaccuracies in word choice and collocation. Additionally, while the essay addresses the drawbacks of city life, it could provide more in-depth analysis on how to overcome these issues.

Overall, the essay shows good language proficiency and effectively presents a well-argued response to the prompt.

Fig. 6. An Example of Feedback Provided by ChatGPT

There are concerns regarding the use of AES systems like Criterion, which are designed to align with large-scale tests, in educational settings. This is due to differences in the writing tasks and underlying constructs, as noted by Condon (2013). However, with AES using GPT, it may be possible to incorporate rubrics that are actually used in instruction. This would enable custom feedback tailored to real classroom settings, which would be a new and innovative approach to providing written corrective feedback. The impact of this adaptive and automated approach is sure to be significant, both in-class and out-of-class and could lead to a reconstruction of how L2 writing is taught and learned.

Although AES using GPT can attain a certain level of accuracy, as demonstrated by this study, it still falls short of achieving perfect agreement with human raters. Therefore, it should be used in conjunction with human evaluation. In essence, AES using GPT can only serve as a supportive tool and cannot replace human raters or classroom teachers. This view is consistent with the longstanding view expressed in the research of AES and AWE (e.g., Attali et al., 2013; Warschauer & Ware, 2006).

It is crucial to acknowledge the limitations of AES with GPT and suggest future research possibilities. To enable comparison with ChatGPT, our study implemented the GPT model without fine-tuning, which still achieved a certain level of accuracy. However, previous research has established that refining the model through fine-tuning (e.g., Sethi & Singh, 2022) and integrating linguistic features, as we did in this study, is likely to enhance its performance. Our study confirmed the significance of lexical sophistication, lexical diversity, fine-grained syntactic dependency measures, and verb-argument construction measures in predicting essay quality, which has been previously reported (Kim et al., 2018; Kyle et al., 2018, 2021; Kyle & Crossley, 2014, 2017, 2018). To further expand this line of research, it would be worthwhile to compare our approach with the current best-performing method in AES research, which combines BERT and linguistic features (Lagakis & Demetriadis, 2021).

Moreover, the results of GPT may vary based on prompt engineering, or how the prompt is formulated. Our study utilized a zero-shot performance method that excluded essay-scoring samples, since including them may alter the accuracy of the model (Mayer et al., 2023). Therefore, it may be beneficial to compare the results to achieve higher accuracy.

Although our study focused on using the GPT-3 text-davinci-003 model for automated essay scoring, it is worth noting that this model has since been superseded by the more powerful and capable GPT-4, which was released on March 14, 2023. As such, future research could explore whether using GPT-4 would result in improved performance and greater efficiency in evaluating student essays.

Additionally, while this study focused on the accuracy and reliability of AI-based automated essay scoring, it is crucial to contemplate the wider educational implications of these technologies. For instance, there may be concerns about the impact of relying on AI-generated scores on student motivation and engagement with writing tasks. Ethical considerations may also arise from using these technologies in educational settings. Future research could explore these and other educational aspects of AI-based automated essay scoring and evaluate their potential benefits and drawbacks for foreign language education.

Lastly, one of the challenges of deep learning models such as GPT is their lack of explainability, resulting in a black box-like nature in which the process by which results are obtained remains obscure. Addressing the aforementioned considerations could improve the interpretability of AES with GPT and further develop the concept of explainable AI (XAI) (Kumar & Boulanger, 2020) for research and practice.

Conclusion

This study aimed to investigate the feasibility of using an AI language model (i.e., GPT) for AES. The findings of this study suggest that GPT has the potential as a promising tool for AES, which has not been explored extensively in the field. Additionally, the use of linguistic features was found to enhance the accuracy of scoring. While further research is needed to fully examine the capabilities and limitations of the approach, the results of this study demonstrate that AES with GPT can be considered as a viable alternative for evaluating and providing feedback on L2 writing.

Recently, the application of GPT in research and practice has emerged, with anticipated further development, especially with the introduction of ChatGPT. However, the use of ChatGPT is currently being evaluated, and some academic journals have cautioned against its irresponsible application (e.g., Nature Editorial, 2023). Despite criticisms from scholars like Chomsky, who denounce ChatGPT as a form of "high-tech plagiarism" (EduKitchen, 2023), it is evident that AI language models are here to stay, and it is our responsibility to educate students on their ethical use because "the future is now" (Pavlik, 2023, p. 9). At the same time, learners, teachers, and researchers can benefit from acquiring effective methods for utilizing GPT in their work, particularly non-native English speakers, who can be linguistically empowered. Language models such as GPT can be best understood as an ever-present, logical assistant, as AI is not currently a substitute for human expertise. With this in mind, we hope that our study has provided preliminary evidence of the potential utility of GPT in AES and paved the way for an era where AI language models and human expertise can coexist and thrive in the field of applied linguistics.

Declaration of Competing Interest

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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In the preparation of this manuscript, we employed ChatGPT (GPT-3.5 & 4) to enhance the clarity and coherence of the language, ensuring it adheres to the standards expected in scholarly journals. While ChatGPT played a role in refining the language, it did not contribute to the generation of any original ideas. The authors alone are responsible for any inaccuracies present in the manuscript. This research was supported by JSPS KAKENHI Grant No. 21H00553.

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